

Задание 20 - Модули ядра Linux

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1) Создаем файл с модулем и Makefile

```
svikari@svikari-VirtualBox:~$ cd lab20
svikari@svikari-VirtualBox:~/lab20$ touch test1.c
svikari@svikari-VirtualBox:~/lab20$ sudo nano test1.c
svikari@svikari-VirtualBox:~/lab20$ touch makefile
svikari@svikari-VirtualBox:~/lab20$ sudo nano makefile
svikari@svikari-VirtualBox:~/lab20$
```

2) Собираем модуль

```
svikari@svikari-VirtualBox:~/lab20$ make
make -C /lib/modules/6.8.0-124-generic/build M=/home/svikari/lab20 modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-124-generic'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04
.3) 12.3.0
You are using: gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
CC [M] /home/svikari/lab20/test1.o
MODPOST /home/svikari/lab20/Module.symvers
CC [M] /home/svikari/lab20/test1.mod.o
LD [M] /home/svikari/lab20/test1.ko
BTF [M] /home/svikari/lab20/test1.ko
Skipping BTF generation for /home/svikari/lab20/test1.ko due to unavailability o
f vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-124-generic'
```

3) Просматриваем через dmesg

```
svikari@svikari-VirtualBox:/var/log$ sudo dmesg | tail -n10
[ 799.172915] audit: type=1400 audit(1781758917.899:5608): apparmor="DENIED" op
eration="open" class="file" profile="snap.snapd-desktop-integration.snapd-deskto
p-integration" name="/home/svikari/.config/user-dirs.dirs" pid=45645 comm="sed"
requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 799.174056] audit: type=1400 audit(1781758917.901:5609): apparmor="DENIED" op
eration="open" class="file" profile="snap.snapd-desktop-integration.snapd-deskto
p-integration" name="/home/svikari/.config/user-dirs.dirs" pid=45646 comm="deskt
op-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 799.177126] audit: type=1400 audit(1781758917.904:5610): apparmor="DENIED" op
eration="open" class="file" profile="snap.snapd-desktop-integration.snapd-deskto
p-integration" name="/home/svikari/.config/user-dirs.locale" pid=45647 comm="cp"
requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 799.178293] audit: type=1400 audit(1781758917.906:5611): apparmor="DENIED" op
eration="open" class="file" profile="snap.snapd-desktop-integration.snapd-deskto
p-integration" name="/home/svikari/.config/user-dirs.locale" pid=45648 comm="des
```

```
[ 1046.355608] audit: type=1400 audit(17817
"snap.snapd-desktop-integration.snapd-deskt
p-i" requested_mask="wr" denied_mask="wr" f
[ 1046.356446] audit: type=1400 audit(17817
"snap.snapd-desktop-integration.snapd-deskt
ktop-i" requested_mask="wr" denied_mask="wr
[ 1048.658368] audit: type=1400 audit(17817
ap.snapd-desktop-integration.snapd-desktop-
ktop-launch" requested_mask="r" denied_mask
[ 1050.910505] My test module loaded!!!!
```

4) Просматриваем модули

```
svikari@svikari-VirtualBox:~/lab20$ lsmod
Module                Size  Used by
test1                  12288  0
snd_intel8x0           53248  2
intel_rapl_msr         20480  0
intel_rapl_common      40960  1 intel_rapl_msr
snd_sof_pci_core       106608  1 snd_intel8x0
```

5) С помощью rmmod удаляем модуль из ядра

```
[ 1262.312346] My test module unloaded!!!!
```

6) Создаем еще модуль с файлами устройств

```
GNU nano 6.2 test3.c *
static char test_string[15] = "Hello!\n\0";

ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
{
    size_t rc;
    read_lock(&lock);
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);
    read_unlock(&lock);
    return rc;
}

ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
{
    size_t rc = 0;
    if (size > 15)
        return -EINVAL;
    write_lock(&lock);
    rc = simple_write_to_buffer(test_string, 15, off, buff, size);
    write_unlock(&lock);
    return rc;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .read = test_read,
    .write = test_write,
}
```

7) Проверяем, что он загружен и смотрим его номер, с помощью которого записываем его в системе

```
[ 2345.297605] Test module is loaded.
[ 2345.297610] Major number is 240.
```

```
svikari@svikari-VirtualBox:/dev$ sudo mknod test3 c 240 0
svikari@svikari-VirtualBox:/dev$ ls -la test3
crw-r--r-- 1 root root 240, 0 jun 18 07:29 test3
svikari@svikari-VirtualBox:/dev$
```

8) Создаем файл test.c для работы с sys и proc

```
svikari@svikari-VirtualBox:~/lab20$ lsmod | grep test
test                12288  0
test3               12288  0
svikari@svikari-VirtualBox:~/lab20$ ls -la /sys/kernel/my_simple/
total 0
drwxr-xr-x  2 root root    0 jun 18 07:56 .
drwxr-xr-x 17 root root    0 jun 18 06:48 ..
-rw-rw-r--  1 root root 4096 jun 18 07:56 value
```

```
svikari@svikari-VirtualBox:/proc$ echo "42" | sudo tee /sys/kernel/my_simple/value
cat /sys/kernel/my_simple/value
42
42
```